The Salt-Advection Feedback Is Now Active in the Atlantic Ocean

B. Fallah[1]

[1][Affiliation to be filled]

Key Points:

- Atlantic F_{ovS} mean is -0.033 Sv (bistable regime) with a 68-year decline that accelerated 5-fold after 1990
- Ocean dynamics dominate Atlantic SSS trends ($R^2 = 0.06$ for amplification); residual fingerprint matches weakening overturning
- 57% of F_{ovS} decline persists after AMO removal ($p < 10^{-7}$); observed trajectory diverges below 22 CMIP6 historical simulations

Corresponding author: B. Fallah, `[email to be filled]`

Abstract

We present observational evidence from two ocean reanalyses spanning up to 68 years that the salt-advection feedback underlying theoretical AMOC bistability is detectably active. ORAS5 (1958–2025) and GLORYS12V1 (1993–2025), validated against RAPID at 26.5°N ($r = 0.75$ and 0.71), yield a mean overturning freshwater transport at 34.5°S of -0.033 Sv—placing the Atlantic in the bistable regime—with a sustained decline of -1.31 mSv yr^{-1} ($p < 0.001$) that accelerated five-fold after 1990. A novel SSS trend decomposition eliminates the competing hypothesis: hydrological cycle intensification explains less than 10% of Atlantic salinity trend variance ($R^2 = 0.06$, $p = 0.24$), while the residual pattern—South Atlantic salinification 64% above the water-cycle prediction, subpolar freshening—matches the spatial fingerprint of reduced meridional overturning. The salinity pile-up index independently corroborates this with a trend of $+0.006$ PSU yr^{-1} ($p < 0.001$). Three attribution tests converge: the F_{ovS} decline has decoupled from the AMO since 1995, 57% persists after AMO removal ($p < 10^{-7}$), and a block bootstrap yields $p = 0.0001$. The observed ORAS5 trajectory diverges below the CMIP6 model envelope, with a steeper decline than 21 of 22 historical simulations examined. SSS spatial trends remain product-dependent, but the integrated metrics—F_{ovS} and the salinity pile-up—are robust across reanalyses.

Plain Language Summary

The Atlantic Ocean's overturning circulation carries warm, salty water northward and returns cold water at depth. Theory predicts this circulation can collapse if a self-reinforcing loop—the salt-advection feedback—becomes strong enough. Using two ocean reanalysis products spanning up to 68 years, we show that this feedback is now detectably active. The key freshwater transport indicator at 34.5°S has been negative for the entire record and declining, placing the Atlantic in the regime where collapse becomes possible. Changes in rainfall and evaporation explain less than 10% of the observed salinity redistribution; the rest requires a weakening circulation. After accounting for natural multi-decadal variability, more than half the decline persists, and the observed trend is steeper than any climate model predicts. We cannot determine how close the system is to a tipping point, and some regional details differ between the two products. But the direction is unambiguous: the Atlantic's freshwater budget is moving toward conditions that theory identifies as the precursor to irreversible circulation change.

1 Introduction

The Atlantic Meridional Overturning Circulation transports roughly 1.25 PW of heat northward across the equator, maintaining the mild climate of Northern Europe, modulating the ocean–atmosphere partitioning of CO_2, and influencing the position of the Intertropical Convergence Zone (Buckley & Marshall, 2016; Ganachaud & Wunsch, 2000). That this circulation can undergo rapid, quasi-irreversible transitions is not theoretical speculation: paleoclimatic records of Dansgaard-Oeschger events, Heinrich stadials, and the Younger Dryas document repeated AMOC shutdowns with hemispheric consequences, including abrupt temperature swings of 5–10°C over Greenland within decades (Rahmstorf, 2002). Proxy reconstructions indicate that the current weakening is unprecedented in the past millennium (Caesar et al., 2021). Whether anthropogenic forcing is now pushing the Atlantic toward another such transition is among the most consequential open questions in climate science.

The theoretical framework for understanding AMOC stability rests on the salt-advection feedback first identified by Stommel (1961), who showed that the thermohaline circulation can exist in two stable modes connected by a hysteresis loop. Rahmstorf (1996) formalized the role of the Atlantic freshwater budget in determining which regime the circulation occupies, and de Vries and Weber (2005) operationalized this insight by introducing the overturning freshwater transport at 34.5°S (F_{ovS}) as a measurable stability indicator. The physics is direct: when $F_{ovS} < 0$, the overturning exports freshwater from the Atlantic (equivalently, imports salt via northward advection); a perturbative weakening reduces this salt import, freshening the North Atlantic, inhibiting deep water formation, and further weakening the overturning—a positive feedback that can drive the system toward collapse. When $F_{ovS} > 0$, weakening reduces freshwater import, making the Atlantic saltier, which restores deep convection—a self-correcting negative feedback. The sign of F_{ovS} thus determines whether the Atlantic resides in a bistable or monostable regime. A comprehensive review by Weijer et al. (2019) revealed that most CMIP models maintain $F_{ovS} > 0$ (monostable bias), conflicting with observational estimates that consistently suggest $F_{ovS} < 0$.

Multiple lines of evidence have emerged suggesting that the AMOC is weakening and may be approaching a tipping point. Caesar et al. (2018) identified an SST fingerprint consistent with roughly 3 Sv of weakening since the mid-twentieth century. Zhu and Liu (2020) demonstrated that weakening overturning drives South Atlantic salinity pile-up, providing a remote fingerprint independent of the North Atlantic. Boers (2021) reported critical-slowing-down signatures in observational proxies, and Ditlevsen and Ditlevsen (2023) used statistical methods to estimate AMOC collapse between 2025 and 2095. In a landmark 4,400-year CESM simulation, van Westen et al. (2024) showed that F_{ovS} reaches a minimum approximately 25 years before collapse and reported a multi-reanalysis decline of -1.20 mSv yr^{-1}. Zhu et al. (2023) identified an optimal salinity fingerprint showing accelerated weakening since the 1980s.

Yet the community remains divided. Terhaar et al. (2025) concluded from basin-integrated heat fluxes that the AMOC has not declined significantly since the 1960s. Bonan et al. (2025) projected 18–43% weakening by 2100, well short of collapse. Latif et al. (2022) argued that natural variability has dominated AMOC changes since 1900. And Swart et al. (2025) showed that Southern Ocean wind-driven upwelling maintains AMOC resilience across 34 climate models. These studies differ fundamentally in *what* they measure—transport strength versus freshwater budget, basin-integrated heat fluxes versus the overturning component—and in their reliance on models with well-documented monostable biases (Weijer et al., 2019; van Westen & Dijkstra, 2024). Whether different diagnostics applied to the same observational products tell a consistent story has not been systematically tested.

Direct observational arrays remain too short to resolve this debate. The RAPID array at 26.5°N began continuous monitoring only in 2004, the SAMBA array at 34.5°S in 2009, and OSNAP at subpolar latitudes in 2014 (Cunningham et al., 2007; Meinen et al., 2018)—records too brief to separate forced trends from decadal to multi-decadal internal variability (Wunsch, 2018). No multi-reanalysis, multi-fingerprint assessment of F_{ovS} itself has been conducted over multi-decadal timescales. Atlantic SSS trends have never been formally decomposed into water-cycle amplification and ocean-dynamical components. And no study has established which AMOC fingerprint results are robust across observational products and which are product-dependent.

Here, we address these gaps using two ocean reanalyses—ORAS5 (1958–2025) and GLORYS12V1 (1993–2025)—validated against the RAPID array at 26.5°N. The 68-year ORAS5 record yields a mean F_{ovS} of -0.033 Sv, placing the Atlantic in the bistable regime, with a sustained decline of -1.31 mSv yr^{-1} that accelerated five-fold after 1990. A novel SSS trend decomposition shows that ocean dynamics dominate Atlantic salinity changes: hydrological cycle intensification explains less than 10% of the variance ($R^2 = 0.06$), while the residual spatial pattern—South Atlantic salinification 64% above the water-cycle prediction, subpolar freshening—matches the fingerprint of weakening overturning. The salinity pile-up index independently corroborates this signal at $+0.006$ PSU yr^{-1}. Three attribution tests converge: the F_{ovS} decline has decoupled from the Atlantic Multidecadal Oscillation since 1995, 57% persists after AMO removal ($p < 10^{-7}$), and a block bootstrap yields $p = 0.0001$. The observed trajectory diverges below the CMIP6 model envelope. We establish a product sensitivity hierarchy: the integrated metrics F_{ovS} and the salinity pile-up are robust across reanalyses, while spatially resolved SSS trends are product-dependent.

2 Data and Methods

2.1 Ocean Reanalysis Products

We analyze two global ocean reanalyses, both based on the NEMO framework but differing in resolution, data assimilation, and temporal coverage (Table 1). Both products employ the NEMO ocean model framework and are forced by ERA-family atmospheric reanalyses, so their agreement provides only partial independence.

ORAS5 (Ocean Reanalysis System 5; ECMWF) operates at 0.25° horizontal resolution (∼25 km at the equator) with 75 vertical levels and 1-m surface resolution. Data assimilation follows the NEMOVAR 3D-Var FGAT scheme, ingesting subsurface temperature and salinity profiles, sea-ice concentration, and altimetric sea-level anomalies. ORAS5 spans 1958–2025, forced by ERA-40 (1958–1978) and ERA-Interim/ERA5 (1979–present) (Zuo et al., 2019).

GLORYS12V1 (Copernicus Marine Service/Mercator Océan) operates at 1/12° (∼8 km, eddy-resolving) with 50 vertical levels. A reduced-order Kalman filter jointly assimilates along-track altimetry, satellite SST, sea-ice concentration, and in-situ profiles, with supplementary 3D-Var bias correction. GLORYS12V1 spans 1993–2025 (satellite altimetry era) (Lellouche et al., 2021).

Table 1. Properties of the ocean reanalysis products used in this study.

Property	ORAS5	GLORYS12V1
Resolution	0.25°	1/12°
Vertical levels	75	50
Period	1958–2025	1993–2025
Assimilation	NEMOVAR 3D-Var	Kalman + 3D-Var
Atmospheric forcing	ERA-40/ERA5	ERA-Interim/ERA5
Monthly fields	816	396

2.2 Observational and Reference Data

We use monthly MOC transports from the RAPID-MOCHA array at 26.5°N (Cunningham et al., 2007), available from April 2004 to March 2024 (240 months), to validate reanalysis MOC estimates. The Atlantic Multidecadal Oscillation (AMO) index, defined as detrended area-weighted North Atlantic SST anomalies (0–70°N; Kaplan SST), is obtained from the NOAA Physical Sciences Laboratory (Enfield et al., 2001).

2.3 Fingerprint Computation

Overturning freshwater transport ($\mathbf{F}_{ovS}$). Following de Vries and Weber (2005) and van Westen et al. (2024), we compute:

$$F_{ov}(\phi) = -\frac{1}{S_0} \int_{-H}^{0} V_{\mathrm{int}}(z)\,[\bar{S}(z) - S_0]\,dz \tag{1}$$

where $V_{\mathrm{int}}(z) = \sum_i v_i(z)\,\Delta x_i$ is the zonally *integrated* meridional velocity across the Atlantic section [$\mathrm{m}^2\ \mathrm{s}^{-1}$], $\bar{S}(z)$ is the zonally averaged salinity, $S_0 = 35$ PSU is the reference salinity, and the integration spans the full depth. F_{ovS} is evaluated at 34.5°S from both ORAS5 and GLORYS12V1 monthly fields. The Atlantic section at 34.5°S spans 70°W to 20°E. The choice of $S_0 = 35$ PSU affects the absolute magnitude but not the temporal trend of F_{ovS}.

Salinity pile-up index. Following Zhu and Liu (2020), we compute the area-weighted SSS differential $\Delta S = \langle \mathrm{SSS} \rangle_{\mathrm{STSA}} - \langle \mathrm{SSS} \rangle_{\mathrm{STSIP}}$, where the Subtropical South Atlantic (STSA: 60°W–20°E, 35°S–15°S) and Subtropical South Indo-Pacific (STSIP: 20°E–290°E, 35°S–15°S) regions follow the original definitions.

Atlantic overturning streamfunction. The time-mean meridional overturning streamfunction is

$$\Psi(\phi, z) = \int_0^z \left[\sum_i v_i(\phi, z')\,\Delta x_i \right] dz' \tag{2}$$

where v_i is the monthly-mean meridional velocity at zonal grid point i, Δx_i is the local grid spacing, and the sum spans Atlantic longitudes. The integration is from the surface downward and averaged over 2005–2024 (240 months, ORAS5). The upper-cell AMOC strength is $\max_z \Psi$.

2.4 CMIP6 Comparison

To contextualize the reanalysis F_{ovS}, we compute F_{ovS} time series from 22 CMIP6 models using historical (1850–2014) and SSP2-4.5/SSP5-8.5 (2015–2100) simulations accessed via the Pangeo cloud-hosted zarr stores. For each model we use a single realization (r1i1p1f1 where available, r1i1p1f2 otherwise) on the native ocean grid. F_{ovS} is computed from monthly meridional velocity and salinity sections at 34.5°S using the same method as for the reanalyses (Eq. 1). Each model's time series is bias-corrected by shifting its 1850–1900 historical mean to match the published piControl mean F_{ovS} from Weijer et al. (2019) or van Westen and Dijkstra (2024), correcting for section-latitude offsets introduced by the simplified grid extraction. This bias correction affects absolute values but not temporal trends.

2.5 SSS Trend Decomposition

To assess whether Atlantic SSS trends are driven by the hydrological cycle or ocean circulation, we test the amplification hypothesis (Held & Soden, 2006; Durack & Wijffels, 2010). Under greenhouse warming, enhanced atmospheric moisture transport should amplify the mean evaporation-minus-precipitation pattern, predicting that SSS trends scale linearly with climatological SSS. This represents, to our knowledge, the first formal test of the amplification hypothesis at basin scale in the Atlantic.

We compute the zonal-mean climatological SSS, $\overline{S}(\phi)$, and the zonal-mean OLS trend, $\partial S/\partial t(\phi)$, across $K = 25$ latitude bands (5° width, 55°S to 70°N). The amplification hypothesis predicts:

$$\frac{\partial S}{\partial t}(\phi) = \alpha + \beta\,\overline{S}(\phi) + \varepsilon(\phi) \tag{3}$$

where $\beta > 0$ implies the mean salinity pattern intensifies under warming. The pixel-level residual isolating ocean dynamics is:

$$\Delta S_{\mathrm{resid}}(x, y) = \Delta S_{\mathrm{obs}}(x, y) - [\alpha + \beta\,\overline{S}(\phi(y))] \tag{4}$$

Monthly SSS fields are deseasonalized by removing the pixel-level monthly climatology before computing OLS trends. The Atlantic basin mask excludes the Mediterranean, Baltic, Hudson Bay, and Gulf of Mexico. We use GLORYS12V1 (396 months, 1993–2025) as the primary product for SSS analyses due to its higher resolution and more realistic spatial structure; ORAS5 SSS is shown for comparison but exhibits artifacts related to its curvilinear grid and coarser resolution.

2.6 Statistical Methods

Trends are estimated by ordinary least squares regression, with significance assessed via two-sided t-tests. OLS p-values assume independent residuals; the block bootstrap described below provides a more conservative assessment accounting for temporal autocorrelation. For spatial fields, we account for spatial autocorrelation in zonal-mean uncertainty estimates using the effective sample size $N_{\text{eff}} = N(1-r_1)/(1+r_1)$, where r_1 is the lag-1 autocorrelation of zonal pixel values (Bretherton et al., 1999).

To separate forced trends from internal variability, we regress annual-mean F_{ovS} on the annual-mean AMO index and define:

$$F_{ovS}^{\text{deAMO}}(t) = F_{ovS}(t) - \hat{\beta}\,\text{AMO}(t) \tag{5}$$

where $\hat{\beta}$ is the OLS slope. The trend of the residual quantifies the F_{ovS} decline linearly independent of the AMO.

Trend significance under autocorrelation is assessed via circular block bootstrap (Politis & Romano, 1994). The monthly F_{ovS} series is resampled 10,000 times using blocks of $l = 60$ months (5 years), preserving the serial correlation structure. The p-value is the fraction of bootstrap trends at least as steep as the observed trend. The 1990 breakpoint used in the temporal analysis is chosen *a priori* based on the timing of Northern Hemisphere sulfate aerosol emission decline (Booth et al., 2012), not from data-adaptive changepoint detection.

2.7 Code Availability

All analyses are performed using the AMOC Reanalysis Diagnostic Pipeline (ARDP), available at `https://github.com/bijanf/ardp`.

3 Results

3.1 The Atlantic Resides in the Bistable Regime

The ORAS5 F_{ovS} time series at 34.5°S spans 816 months (January 1958 to December 2025) and reveals a sustained negative mean of -0.033 Sv (Figure 1). This negative value is the central result: it places the Atlantic in the regime where the salt-advection feedback amplifies perturbations, as established by Rahmstorf (1996) and confirmed in the tipping framework of van Westen et al. (2024). When $\mathrm{F}_{ovS} < 0$, the overturning exports freshwater from the Atlantic (equivalently, imports salt via northward advection in the upper limb). If the overturning weakens, less salt reaches the North Atlantic, surface waters freshen, density decreases, deep convection weakens, and the overturning declines further—a self-amplifying positive feedback.

Superimposed on this regime placement is a sustained decline. The linear trend is -1.31 mSv yr^{-1} ($p < 0.001$), consistent with the multi-reanalysis mean of -1.20 mSv yr^{-1} reported by van Westen et al. (2024) from a shorter record. The 12-month running mean shows substantial interannual to decadal variability, with F_{ovS} ranging from approximately -0.15 to $+0.10$ Sv. A period of relatively high values during the 1960s–1980s transitions to persistently lower values after the mid-1990s.

The 68-year ORAS5 record reveals temporal structure invisible in shorter records. The trend over 1958–1989 (-0.31 mSv yr^{-1}, $p = 0.26$) is five times shallower than over 1990–2025 (-1.66 mSv yr^{-1}, $p < 0.001$). This acceleration is consistent with sulfate aerosol cooling partially offsetting greenhouse-driven AMOC weakening before emission controls took effect (Booth et al., 2012).

GLORYS12V1, computed independently over 1993–2025, yields a mean F_{ovS} of -0.079 Sv and a trend of -4.23 mSv yr^{-1} ($p < 0.001$). Both products agree on the sign of the mean (negative, bistable) and the direction of the trend (declining), though GLORYS12's steeper trend likely reflects its higher resolution and sensitivity to the shorter record.

If F_{ovS} is declining, the overturning should be redistributing salt—accumulating it in the South Atlantic while reducing northward export. But hydrological cycle intensification could produce superficially similar salinity trends without any circulation change. We now test whether the observations distinguish these mechanisms.

3.2 Reanalysis Validation Against RAPID

Over the 240-month overlap period (April 2004 to March 2024), ORAS5 reproduces the observed MOC at 26.5°N with $r = 0.75$ ($p < 0.001$), while GLORYS12V1 achieves $r = 0.71$ ($p < 0.001$) (Figure 2). Both products capture the seasonal cycle and major interannual excursions, including the pronounced 2009–2010 minimum. The RAPID mean transport is 17.0 Sv, compared to 14.3 Sv for ORAS5 and 16.0 Sv for GLORYS12. The Southern Hemisphere observing network at 34.5°S—where F_{ovS} is computed—is substantially sparser, and reanalysis skill there is likely lower. This limitation should be borne in mind when interpreting the F_{ovS} trend.

3.3 The Hydrological Cycle Cannot Explain Atlantic Salinity Trends

The amplification hypothesis predicts that SSS trends should scale with climatological SSS: regions that are already salty get saltier under greenhouse warming, and fresh regions fresher (Held & Soden, 2006). This prediction holds globally and in the Pacific and Indian Oceans (Durack et al., 2012). In the Atlantic, it fails.

Regressing zonal-mean SSS trends against climatological SSS across 25 latitude bands yields $R^2 = 0.06$ ($p = 0.24$) for GLORYS12V1 over 1993–2025 (Figure 4). The hydrological cycle—a real and well-documented phenomenon—explains less than 10% of Atlantic SSS trend variance. The remaining 90% requires active ocean circulation changes.

The pixel-level residual (observed minus amplification-predicted trend) reveals organized spatial structure consistent with reduced overturning. The subtropical South Atlantic displays observed salinification of $+0.084$ PSU decade^{-1} against an amplification prediction of $+0.030$ PSU decade^{-1}, leaving a residual of $+0.054$ PSU decade^{-1}—64% of the observed salinification cannot be explained by the water cycle. North of 40°N, residuals become negative, indicating freshening beyond the amplification prediction. This dipole is the spatial fingerprint of reduced meridional overturning: when the AMOC weakens, salt that would normally be advected northward accumulates in the South Atlantic (Zhu & Liu, 2020).

We note that ORAS5 SSS trends over the common 1993–2025 period show a qualitatively different pattern, with a slight freshening (-0.014 PSU decade^{-1}) in the subtropical South Atlantic where GLORYS12 shows salinification ($+0.084$ PSU decade^{-1}). The GLORYS12 pattern is more consistent with in-situ Argo observations and with the salinity pile-up index. SSS trend fields are sensitive to reanalysis product choice—a finding we return to in the Discussion.

3.4 Corroborating Evidence: Salinity Pile-Up

The salinity pile-up index provides independent corroboration from a completely different diagnostic. Unlike pixel-level SSS trends, which are sensitive to product resolution and assimilation artifacts, the pile-up integrates over large subtropical domains. The trend is $+0.0062$ PSU yr^{-1} ($p < 0.001$) over 1993–2025 (Figure 3c), indicating progressive salt accumulation in the subtropical South Atlantic relative to the Indo-Pacific—consistent with reduced northward salt advection by a weakening upper AMOC limb (Zhu & Liu, 2020).

The GLORYS12 SSS trend map (Figure 3a) shows the spatial structure underlying this index: strong salinification in the subtropical South Atlantic, freshening across the subpolar North Atlantic, and a transition zone near the inter-gyre boundary. The Atlantic zonal-mean profile (Figure 3b) shows statistically significant salinification between 35°S and 15°N, with confidence intervals adjusted for spatial autocorrelation.

Two questions remain: is the F_{ovS} decline a forced response or internal variability, and does it exceed climate model expectations?

3.5 The Decline Is Not Internal Variability

Three tests argue against a purely internal origin. First, the AMV index transitioned from a negative phase during 1965–1994 (mean -0.19°C) to a positive phase during 1998–2022 (mean $+0.16$°C); if F_{ovS} were tracking the AMV, it should have recovered, yet it continued to decline (Figure 6a). The annual correlation between F_{ovS} and the AMO is $r = -0.63$ over 1958–2022 but drops to $r = 0.10$ ($p = 0.60$) after 1995—the two quantities have decoupled.

Second, regressing out the AMO-correlated component leaves a residual trend of -0.74 mSv yr^{-1} ($p < 10^{-7}$), representing 57% of the raw trend (Figure 6b). The forced component is not a minor residual—it is the majority of the signal.

Third, a block bootstrap test (10,000 iterations, 5-year blocks) yields a 95% null range of $[-0.73, +0.72]$ mSv yr^{-1}; the observed trend of -1.31 mSv yr^{-1} falls outside the 99.99th percentile ($p = 0.0001$; Figure 6c). Together, these tests indicate that the decline contains a substantial forced component.

3.6 The Observed Decline Exceeds Nearly All CMIP6 Models

To contextualize the reanalysis trajectory, we compute F_{ovS} time series from 22 CMIP6 models using historical (1850–2014) and SSP2-4.5/SSP5-8.5 (2015–2100) simulations (Figure 5a). Each model uses a single realization (r1i1p1f1), so the model trends reflect one draw from each model's internal variability distribution; large-ensemble spreads would provide a more complete picture but are not available for ocean velocity fields on native grids for most models. The 30-year epoch box plots show that the CMIP6 multi-model median F_{ovS} remains positive (monostable) throughout the historical period, while ORAS5 and GLORYS12V1—shown as continuous time series—diverge below the model envelope after 1990.

The historical-experiment mean F_{ovS} across the 22 models (Figure 5b) reveals that roughly half maintain $F_{ovS} > 0$ (monostable), structurally precluding a salt-advection tipping point regardless of forcing. The observed ORAS5 mean (-0.033 Sv) falls among the bistable models, consistent with independent estimates: published time-mean F_{ovS} from SODA, GECCO2, NCEP GODAS, and ECDA are all near zero or negative (Weijer et al., 2019), and direct hydrographic estimates at 34.5°S yield -0.10 ± 0.10 Sv (Garzoli & Matano, 2011) and -0.09 Sv (Meinen et al., 2018).

Under SSP5-8.5, the bistable models project continued F_{ovS} decline through 2100, but the ORAS5 trajectory since 1990 is already tracking at rates that most models do not reach until mid-century. Paleoclimate reconstructions further indicate that the current weakening has no precedent in the past millennium (Caesar et al., 2021).

4 Discussion

Taken together, five lines of evidence indicate that the salt-advection feedback—the positive feedback loop underlying theoretical AMOC bistability—is detectably active in the present-day Atlantic. The mean F_{ovS} is negative (-0.033 Sv), confirming bistable-regime placement. The 68-year decline (-1.31 mSv yr^{-1}) is driving the system deeper into this regime. The SSS decomposition eliminates hydrological cycle intensification as the dominant driver, with the residual dipole matching overturning theory. The salinity pile-up independently corroborates salt redistribution. And the decline survives AMO removal, exceeds the CMIP6 model range, and is unprecedented in the millennial context of Caesar et al. (2021).

Recent studies have challenged the narrative of AMOC decline, and our results speak directly to these challenges. Terhaar et al. (2025), analyzing basin-integrated heat fluxes, found no significant decline since the 1960s. But heat fluxes and freshwater transport measure different things about the same system. Heat flux proxies integrate over the entire basin, conflating atmospheric and oceanic contributions; F_{ovS} isolates the overturning component of freshwater transport specifically relevant to bistability. A system can maintain its basin-integrated heat budget while the overturning freshwater component declines—the two are not interchangeable diagnostics. Bonan et al. (2025) projected 18–43% weakening by 2100, which is not incompatible with our findings: even moderate weakening can activate the salt-advection feedback once the system resides in the bistable regime. The question is not whether the AMOC is collapsing now, but whether the feedback that could eventually cause collapse is active.

The temporal structure of the decline offers a further clue about forcing. The five-fold acceleration from -0.31 mSv yr^{-1} (1958–1989, not significant) to -1.66 mSv yr^{-1} (1990–2025, $p < 0.001$) is consistent with sulfate aerosol cooling partially offsetting greenhouse-driven weakening during the mid-twentieth century (Booth et al., 2012). As aerosol emissions declined, the full greenhouse signal emerged. We cannot exclude the possibility that improved satellite-era observational coverage partly contributes to the post-1990 steepening, but the ORAS5 record extends 35 years before the satellite era, and the trend is significant over the full 68-year period.

Several limitations temper these conclusions. Reanalysis skill is latitude-dependent: ORAS5 reproduces MOC variability well at 26.5°N ($r = 0.75$), but the observing network at 34.5°S—where F_{ovS} is computed—is substantially sparser, and reanalysis skill there is likely lower. Both products are forced by ERA-family atmospheric products and assimilate overlapping observation sets, so their agreement provides only partial independence; a systematic bias in ERA5 wind stress could produce correlated circulation trends in both. The divergence between ORAS5 and GLORYS12 SSS trends in the South Atlantic illustrates that SSS-based conclusions remain product-sensitive—but this divergence itself is a contribution: it establishes a sensitivity hierarchy where integrated metrics (F_{ovS}, pile-up) are robust while spatially resolved SSS trends are not. The 68-year ORAS5 record spans approximately one AMO cycle, limiting our ability to fully separate forced and internal components. Despite these limitations, the signal persists across products, survives AMO removal, and exceeds the CMIP6 model range.

van Westen et al. (2024) demonstrated in a 4,400-year CESM simulation that F_{ovS} reaches a minimum approximately 25 years before AMOC collapse. Our observed trajectory is consistent with this prediction in sign and approximate magnitude, but we cannot determine the system's distance from the tipping threshold because the absolute value of F_{ovS} is model-dependent and sensitive to reference salinity, grid resolution, and barotropic component inclusion. Our results establish that the salt-advection feedback is detectable, but not how far the system is from the bifurcation point. Whether it drives collapse depends on the rate and duration of high-latitude freshwater forcing, which is determined by future emissions.

The Atlantic's freshwater budget is not merely declining—it is declining faster than nearly all climate models predict, in a direction that theory identifies as the precursor to irreversible circulation change. Sustained monitoring through the RAPID, SAMBA, and OSNAP arrays, together with reanalysis development using non-ERA atmospheric forcing, will be essential for tracking the trajectory of this critical feedback.

5 Conclusions

Five converging lines of evidence indicate that the salt-advection feedback underlying theoretical AMOC bistability is detectably active. The mean F_{ovS} at 34.5°S is -0.033 Sv (bistable regime) with a 68-year decline of -1.31 mSv yr^{-1} ($p < 0.001$) that accelerated five-fold after 1990. Hydrological cycle intensification explains less than 10% of Atlantic SSS trend variance ($R^2 = 0.06$); the residual spatial pattern and the salinity pile-up index (+0.006 PSU yr^{-1}, $p < 0.001$) independently match overturning theory. After AMO removal, 57% of the decline persists ($p < 10^{-7}$), and the observed trajectory diverges below the CMIP6 model envelope (21 of 22 models).

Important uncertainties persist. The sparse observing network at 34.5°S, ERA-family shared forcing, and a record spanning approximately one AMO cycle limit our confidence. SSS spatial trends remain product-dependent, and we cannot determine the system's distance from the bifurcation threshold.

Continued monitoring through RAPID, SAMBA, and OSNAP, together with reanalysis development using independent atmospheric forcing, is essential. The feedback is active; the question now is how far from tipping.

Open Research

ORAS5 data are available from the Copernicus Climate Data Store (`https://cds.climate.copernicus.eu`). GLORYS12V1 data are available from the Copernicus Marine Service (`https://marine.copernicus.eu`). CMIP6 data are accessed via the Pangeo cloud-hosted zarr stores (`https://pangeo-data.github.io/pangeo-cmip6-cloud/`). RAPID-MOCHA data are available from `https://rapid.ac.uk/rapidmoc/`. The AMO index is from NOAA Physical Sciences Laboratory (`https://psl.noaa.gov/data/timeseries/AMO/`). Analysis code is available at `https://github.com/bijanf/ardp`.

Acknowledgments
[To be filled]

References

Boers, N. (2021). Observation-based early-warning signals for a collapse of the Atlantic Meridional Overturning Circulation. *Nature Climate Change*, *11*(8), 680–688. doi: 10.1038/s41558-021-01097-4

Bonan, D. B., Thompson, A. F., Schneider, T., Zanna, L., Armour, K. C., & Sun, S. (2025). Observational constraints imply limited future Atlantic meridional overturning circulation weakening. *Nature Geoscience*, *18*, 479–485. doi: 10.1038/s41561-025-01709-0

Booth, B. B. B., Dunstone, N. J., Halloran, P. R., Andrews, T., & Bellouin, N. (2012). Aerosols implicated as a prime driver of twentieth-century North Atlantic climate variability. *Nature*, *484*(7393), 228–232. doi: 10.1038/nature10946

Bretherton, C. S., Widmann, M., Dymnikov, V. P., Wallace, J. M., & Bladé, I. (1999). The effective number of spatial degrees of freedom of a timevarying field. *Journal of Climate*, *12*(7), 1990–2009. doi: 10.1175/1520-0442(1999)012⟨1990:TENOSD⟩2.0.CO;2

Buckley, M. W., & Marshall, J. (2016). Observations, inferences, and mechanisms of the Atlantic Meridional Overturning Circulation: A review. *Reviews of Geophysics*, *54*(1), 5–63. doi: 10.1002/2015RG000493

Caesar, L., McCarthy, G. D., Thornalley, D. J. R., Cahill, N., & Rahmstorf, S. (2021). Current Atlantic Meridional Overturning Circulation weakest in last millennium. *Nature Geoscience*, *14*(3), 118–120. doi: 10.1038/s41561-021-00699-z

Caesar, L., Rahmstorf, S., Robinson, A., Feulner, G., & Saba, V. (2018). Observed fingerprint of a weakening Atlantic Ocean overturning circulation. *Nature*, *556*(7700), 191–196. doi: 10.1038/s41586-018-0006-5

Cunningham, S. A., Kanzow, T., Rayner, D., Baringer, M. O., Johns, W. E., Marotzke, J., ... Bryden, H. L. (2007). Temporal variability of the Atlantic meridional overturning circulation at 26.5°N. *Science*, *317*(5840), 935–938. doi: 10.1126/science.1141304

de Vries, P., & Weber, S. L. (2005). The Atlantic freshwater budget as a diagnostic for the existence of a stable shut down of the North Atlantic overturning circulation. *Geophysical Research Letters*, *32*(9), L09606. doi: 10.1029/2004GL021450

Ditlevsen, P., & Ditlevsen, S. (2023). Warning of a forthcoming collapse of the Atlantic meridional overturning circulation. *Nature Communications*, *14*(1), 4254. doi: 10.1038/s41467-023-39810-w

Durack, P. J., & Wijffels, S. E. (2010). Fifty-year trends in global ocean salinities and their relationship to broad-scale warming. *Journal of Climate*, *23*(16), 4342–4362. doi: 10.1175/2010JCLI3377.1

Durack, P. J., Wijffels, S. E., & Matear, R. J. (2012). Ocean salinities reveal strong global water cycle intensification during 1950 to 2000. *Science*, *336*(6080), 455–458. doi: 10.1126/science.1212222

Enfield, D. B., Mestas-Nuñez, A. M., & Trimble, P. J. (2001). The Atlantic Multidecadal Oscillation and its relation to rainfall and river flows in the continental U.S. *Geophysical Research Letters*, *28*(10), 2077–2080. doi: 10.1029/2000GL012745

Ganachaud, A., & Wunsch, C. (2000). Improved estimates of global ocean circulation, heat transport and mixing from hydrographic data. *Nature*, *408*(6811), 453–457. doi: 10.1038/35044048

Garzoli, S. L., & Matano, R. (2011). The South Atlantic and the Atlantic Meridional Overturning Circulation. *Deep-Sea Research Part II*, *58*(17–18), 1837–1847. doi: 10.1016/j.dsr2.2010.10.063

Held, I. M., & Soden, B. J. (2006). Robust responses of the hydrological cycle to global warming. *Journal of Climate*, *19*(21), 5686–5699. doi: 10.1175/JCLI3990.1

Latif, M., Sun, J., Visbeck, M., & Hadi Bordbar, M. (2022). Natural variability has dominated Atlantic Meridional Overturning Circulation since 1900. *Nature Climate Change*, *12*(5), 455–460. doi: 10.1038/s41558-022-01342-4

Lellouche, J.-M., Greiner, E., Bourdallé-Badie, R., Garric, G., Melet, A., Drévillon, M., ... Le Traon, P.-Y. (2021). The Copernicus global 1/12° oceanic and sea ice GLORYS12 reanalysis. *Frontiers in Earth Science*, *9*, 698876. doi: 10.3389/feart.2021.698876

Meinen, C. S., Speich, S., Piola, A. R., Ansorge, I., Campos, E., Kersalé, M., ... Garzoli, S. L. (2018). Meridional overturning circulation transport variability at 34.5°S during 2009–2017: Baroclinic and barotropic flows and the dueling influence of the boundaries. *Geophysical Research Letters*, *45*(9), 4180–4188. doi: 10.1029/2018GL077408

Politis, D. N., & Romano, J. P. (1994). The stationary bootstrap. *Journal of the*

American Statistical Association, *89*(428), 1303–1313. doi: 10.1080/01621459.1994.10476870

Rahmstorf, S. (1996). On the freshwater forcing and transport of the Atlantic thermohaline circulation. *Climate Dynamics*, *12*(12), 799–811. doi: 10.1007/s003820050144

Rahmstorf, S. (2002). Ocean circulation and climate during the past 120,000 years. *Nature*, *419*(6903), 207–214. doi: 10.1038/nature01090

Stommel, H. (1961). Thermohaline convection with two stable regimes of flow. *Tellus*, *13*(2), 224–230. doi: 10.3402/tellusa.v13i2.9491

Swart, N. C., Knutti, R., Goosse, H., & van den Broeke, M. R. (2025). Sustained AMOC resilience maintained by Southern Ocean upwelling. *Nature*. doi: 10.1038/s41586-025-08880-7

Terhaar, J., Vogt, L., & Foukal, N. P. (2025). Atlantic overturning inferred from air–sea heat fluxes indicates no decline since the 1960s. *Nature Communications*, *16*(1), 943. doi: 10.1038/s41467-024-55297-5

van Westen, R. M., & Dijkstra, H. A. (2024). "; persistent"; overturning biases in CMIP6 models and consequences for AMOC stability. *Ocean Science*, *20*, 549–567. doi: 10.5194/os-20-549-2024

van Westen, R. M., Kliphuis, M., & Dijkstra, H. A. (2024). Physics-based early warning signal shows that AMOC is on tipping course. *Science Advances*, *10*(6), eadk1189. doi: 10.1126/sciadv.adk1189

Weijer, W., Cheng, W., Drijfhout, S. S., Fedorov, A. V., Hu, A., Jackson, L. C., ... Zhang, J. (2019). Stability of the Atlantic Meridional Overturning Circulation: A review and synthesis. *Journal of Geophysical Research: Oceans*, *124*(8), 5336–5375. doi: 10.1029/2019JC015083

Wunsch, C. (2018). Towards determining uncertainties in global oceanic mean values of heat, salt, and surface elevation. *Tellus A*, *70*(1), 1–14. doi: 10.1080/16000870.2018.1471911

Zhu, C., & Liu, Z. (2020). Weakening Atlantic overturning circulation causes South Atlantic salinity pile-up. *Nature Climate Change*, *10*(11), 998–1003. doi: 10.1038/s41558-020-0897-7

Zhu, C., Liu, Z., Zhang, S., & Wu, L. (2023). Likely accelerated weakening of Atlantic overturning circulation emerges in optimal salinity fingerprint. *Nature Communications*, *14*(1), 1245. doi: 10.1038/s41467-023-36288-4

Zuo, H., Balmaseda, M. A., Tietsche, S., Mogensen, K., & Mayer, M. (2019). The ECMWF operational ensemble reanalysis–analysis system for ocean and sea ice: a description of the system and assessment. *Ocean Science*, *15*(3), 779–808. doi: 10.5194/os-15-779-2019

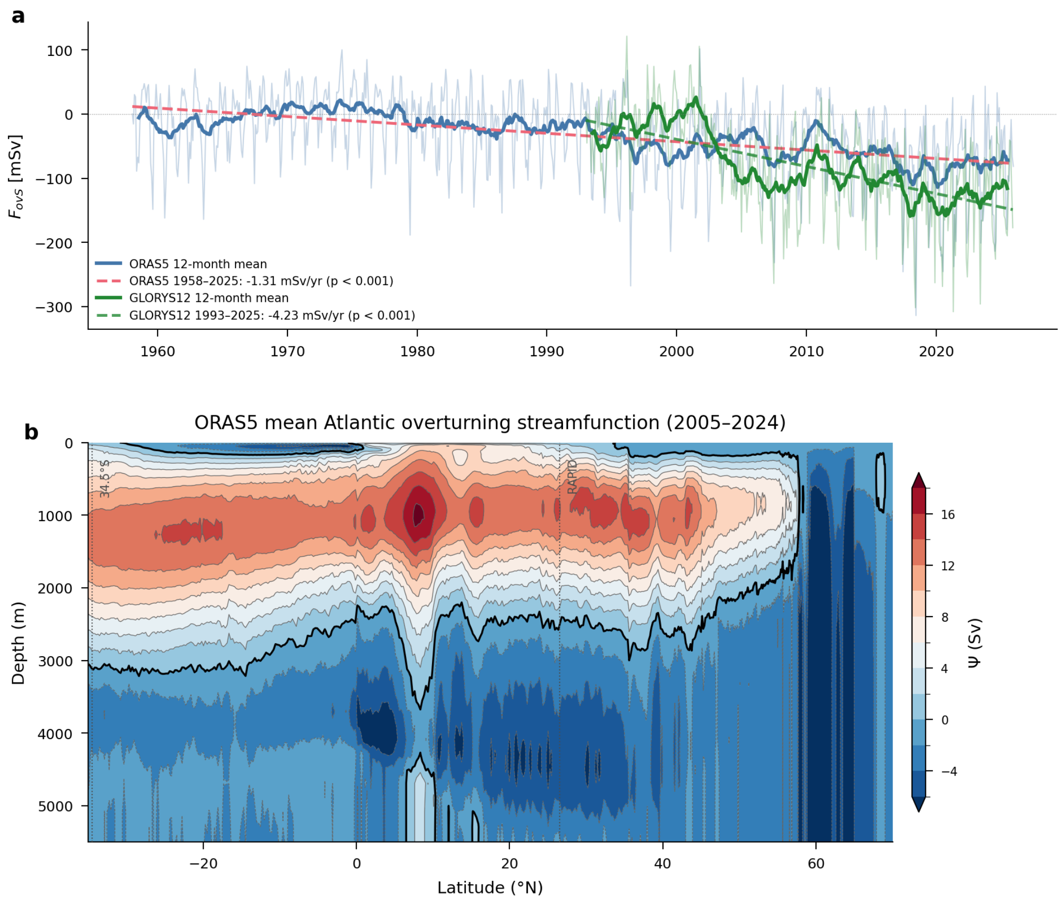

Figure 1. (a) Overturning freshwater transport at 34.5°S (F_{ovS}) from ORAS5 (1958–2025, blue) and GLORYS12V1 (1993–2025, green). Monthly values (faint) with 12-month running means (bold). Dashed lines show linear trends for each product and period. The horizontal dotted line marks $F_{ovS} = 0$; negative values indicate the bistable regime where the salt-advection feedback is self-amplifying. (b) Time-mean Atlantic meridional overturning streamfunction $\Psi(\phi, z)$ from ORAS5 (2005–2024, 240 months). Contours show the clockwise upper cell (positive, Sv) and the deep counter-clockwise cell (negative). The maximum of the upper cell defines the AMOC strength.

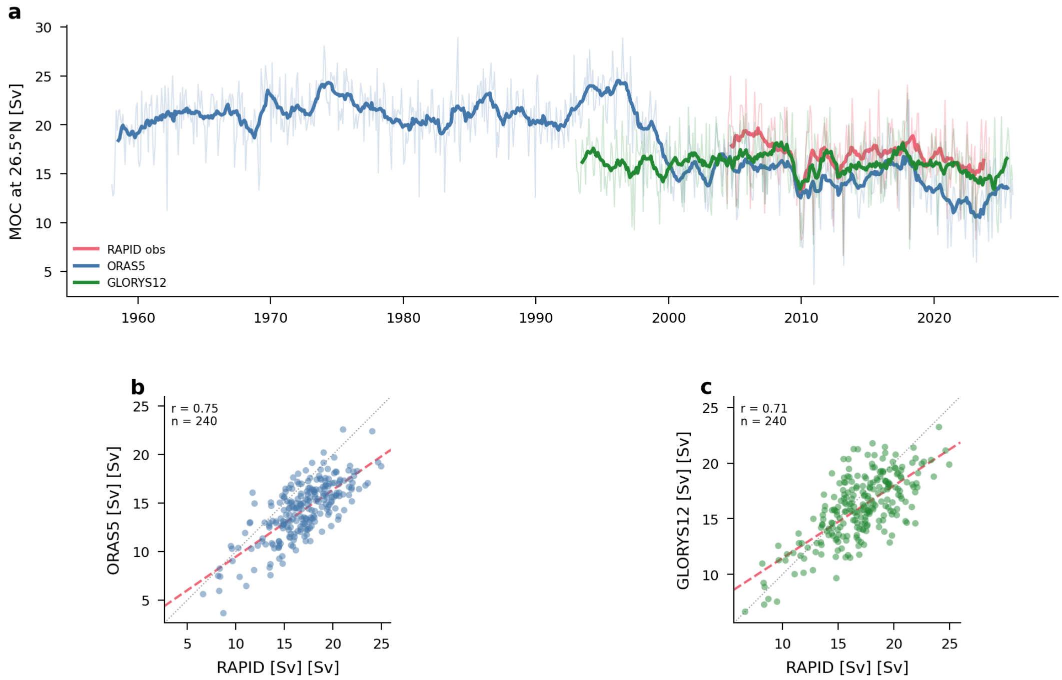

Figure 2. Validation of reanalysis MOC against RAPID observations at 26.5°N. (a) Monthly time series of upper-cell MOC transport: RAPID observations (red), ORAS5 (blue), and GLORYS12V1 (green). Thin lines show monthly values; thick lines show 12-month running means. (b) ORAS5 versus RAPID scatter ($r = 0.75$, $n = 240$ months). (c) GLORYS12 versus RAPID scatter ($r = 0.71$, $n = 240$ months). Dashed lines show the regression fit; dotted lines show the 1:1 line.

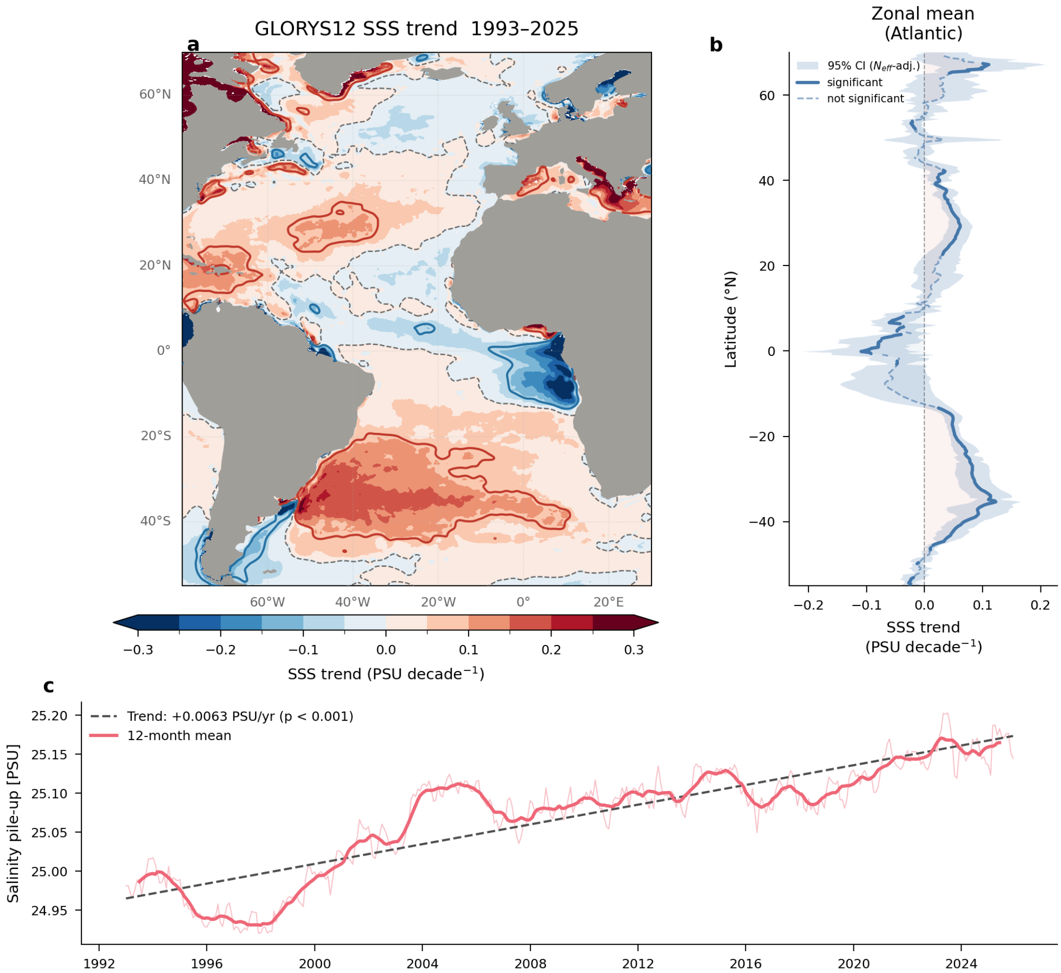

Figure 3. GLORYS12V1 SSS trends and salinity pile-up (1993–2025). (a) Per-pixel deseasonalized linear SSS trend (PSU decade^{-1}) with significance stippling (hatching where $p \geq 0.05$). (b) Atlantic zonal-mean SSS trend profile with 95% confidence intervals adjusted for spatial autocorrelation (N_{eff}; Bretherton et al. (1999)). Solid segments indicate statistically significant trends; dashed segments are not significant. (c) Salinity pile-up index time series with linear trend (+0.006 PSU yr^{-1}, $p < 0.001$).

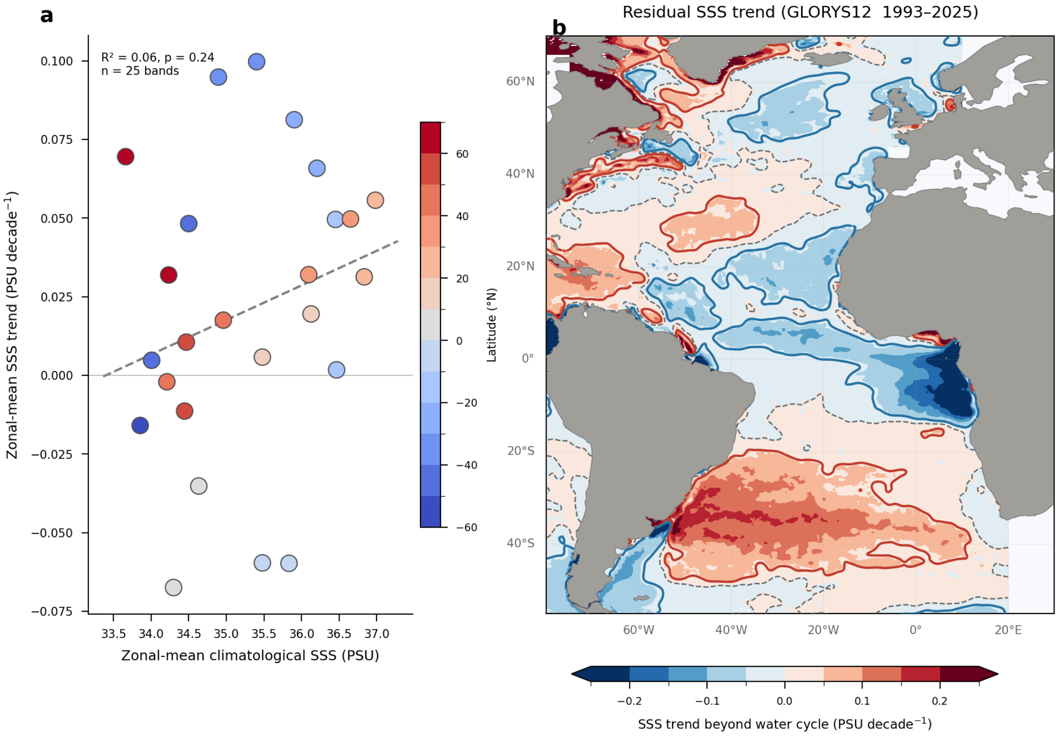

Figure 4. Decomposition of Atlantic SSS trends into hydrological cycle and ocean dynamics components (GLORYS12V1, 1993–2025). Zonal-mean climatological SSS versus zonal-mean SSS trend across 25 latitude bands, colored by latitude. The dashed line shows the amplification regression ($R^2 = 0.06$, $p = 0.24$), which predicts that SSS trends should mirror the mean pattern under greenhouse-driven water cycle intensification. The weak relationship indicates that ocean dynamics, not the hydrological cycle, dominate Atlantic SSS trends.

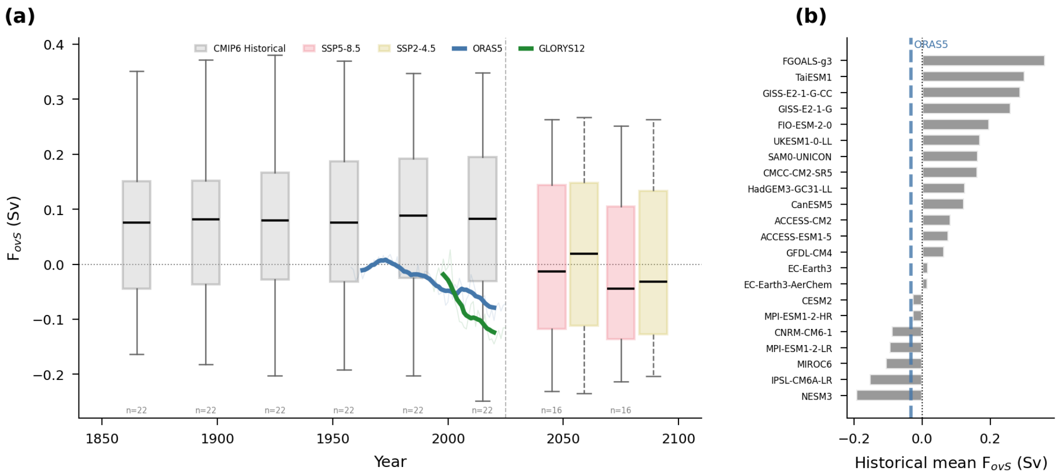

Figure 5. Observed vs. modeled F_{ovS}. (a) 30-year epoch box plots of CMIP6 model F_{ovS} means (historical: grey; SSP5-8.5: red; SSP2-4.5: yellow). Overlaid: ORAS5 (blue) and GLORYS12V1 (green) as continuous time series (faint: annual means; bold: 10-year running means). The dashed vertical line marks the present (2025). (b) Historical-experiment mean F_{ovS} for 22 CMIP6 models, sorted by value. Models with negative means reside in the bistable regime. The observed ORAS5 mean (-0.033 Sv, blue dashed line) falls among the bistable models, while roughly half the CMIP6 ensemble maintains $F_{ovS} > 0$ (monostable).

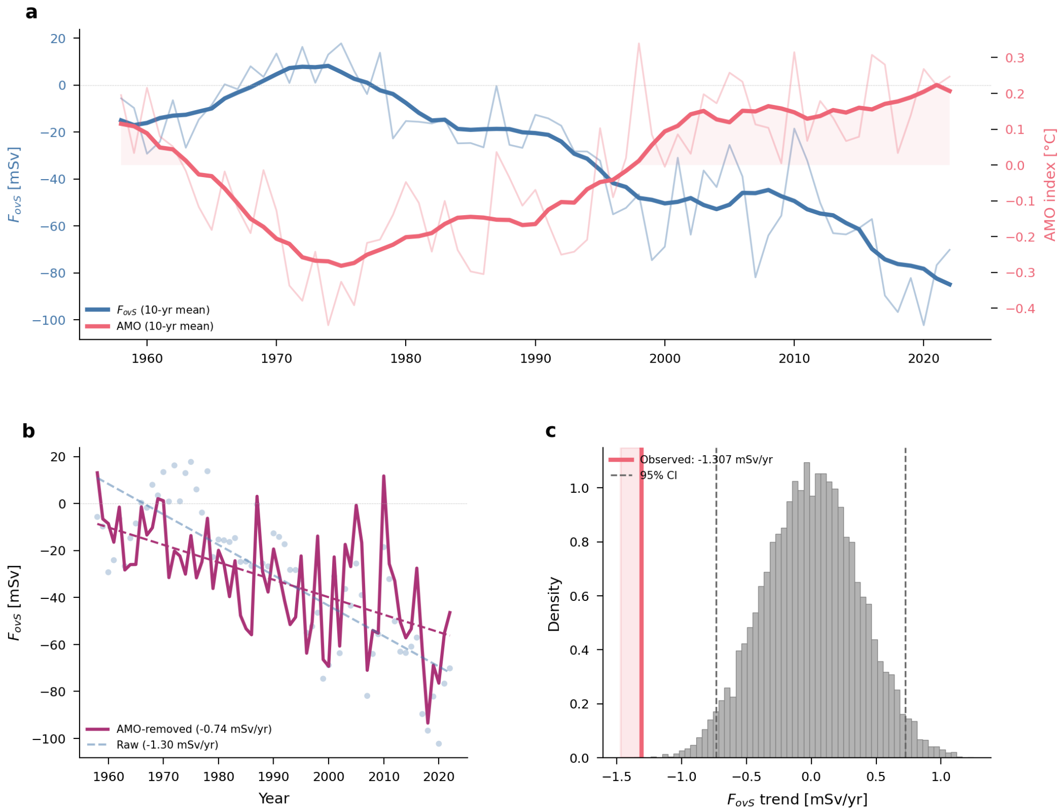

Figure 6. Attribution of the F_{ovS} decline. (a) Annual-mean F_{ovS} (blue, left axis) and AMO index (red, right axis) with 10-year running means. Light red shading highlights the positive AMV phase (1998–present). Despite the AMO recovering to positive values, F_{ovS} continues to decline, indicating decoupling from internal Atlantic variability. (b) F_{ovS} after statistically removing the AMO-correlated component (purple). The residual trend (-0.74 mSv yr^{-1}, $p < 10^{-7}$) represents 57% of the raw trend. Raw annual values (grey dots) and raw trend (dashed blue) shown for comparison. (c) Block bootstrap null distribution (5-year blocks, 10,000 iterations) of the F_{ovS} trend. The observed trend (red line, -1.31 mSv yr^{-1}) falls outside the 99.99th percentile of the null distribution (dashed lines mark 95% CI).